Curs 10 – Regression Models

Simple Linear Regression Models

Overview

1. Definition of a good model
2. Estimation of model parameters
3. Allocation of variation
4. Standard deviation of errors
5. Confidence intervals for regression parameters
6. Confidence intervals for predictions
7. Visual tests for verifying regression assumption

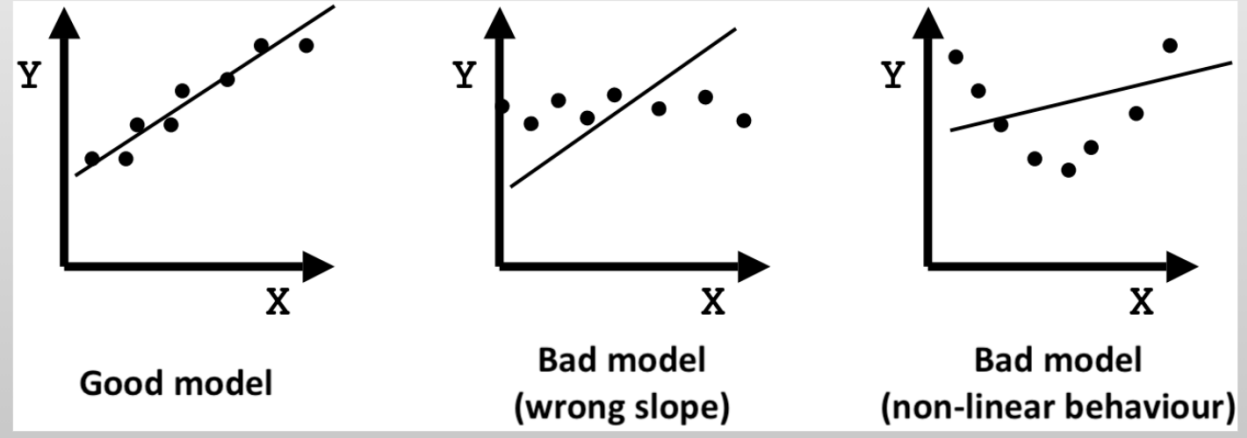
Simple linear regression models

- Regression model
    - Predict a response for a given set of predictor variables
    - Response variable
        - Estimated variable
    - Predictor variables
        - Variables used to predict the response predictors or factors
- Linear regression models
    - Response is a linear function of predictors
- Simple linear regression models
    - Only one predictor

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Good Model



- Regression models attempt to minimize the distance measured vertically between the observation point and the model line
- The length of the line segment is called residual / modelling error / error
- The negative and positive errors should cancel out
  - => zero overall error
  - Many lines satisfy this criterion
- Choose the line that minimizes the sum of squares of the errors

$$\widehat{y} = b_0 + b_1 x$$

  - $\hat{y}$ is the predicted response when the predictor variable is *x*
  - b0 and b1 are fixed regression parameters to be determined from the data
- Given n observation pairs { (x1, y1), …, (xn, yn)}, the estimated response $\hat{y}_i$ for the ith observation is

$$\widehat{y_i} = b_0 + b_1 x_i$$

- The error is

$$e_i = y_i - \widehat{y_i}$$

- The best linear model minimizes the sum of squared errors (SSE)

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right)^2$$

- Subject to the constraint that the mean error is zero

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right) = 0$$

- This is equivalent to minimizing the variance of errors

Estimation of mode parameters

- Regression parameters that give minimum error variance

$$b_1 = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n\overline{x}^2} \qquad \text{and} \qquad b_0 = \overline{y} - b_1\overline{x}$$

- Where

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\sum xy = \sum_{i=1}^{n} x_i y_i \qquad \sum x^2 = \sum_{i=1}^{n} x_i^2$$

Example

- The number of disk I/O's and processor times of seven programs were measured as {(14, 2), (16, 5), (27, 7 (42, 9), (39, 10), (50, 13), (83, 20)}.
- For this data

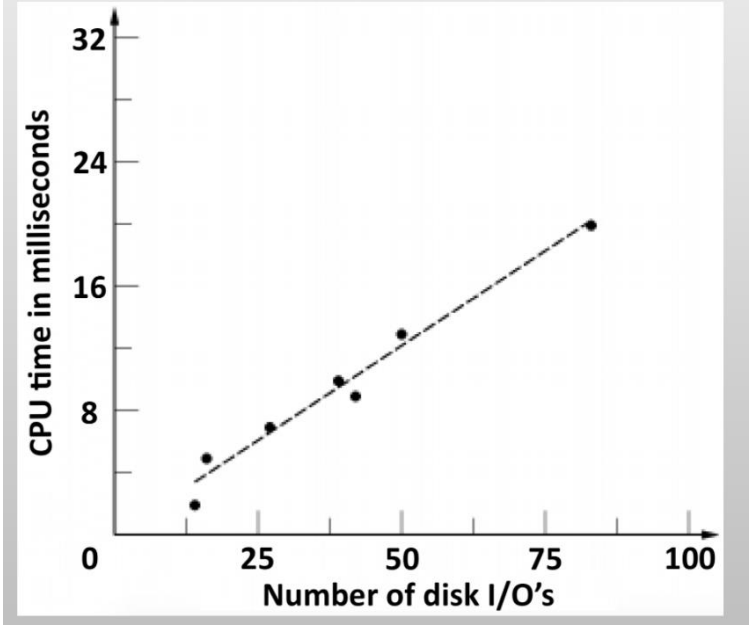
$$n = 7,\ \sum xy = 3375,\ \sum x = 271,\ \sum x^2 = 13.855$$
$$\sum y = 66,\ \sum y^2 = 828,\ \overline{x} = 38.71,\ \overline{y} = 9.43$$

$$b_1 = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n(\overline{x})^2} = \frac{3375 - 7 \times 38.71 \times 9.43}{13.855 - 7 \times (38.71)^2} = 0.2438$$

$$b_0 = \overline{y} - b_1\overline{x} = 9.43 - 0.2438 \times 38.71 = -0.0083$$

- The desired linear model is:

$$CPU\,time = -0.0083 + 0.2438(Number\,of\,disk\,I/O's)$$



	Disk I/O's	CPU Time	Estimate	Error	Error^2
	x_i	y_i	$\widehat{y}_i = b_0 + b_1 x_i$	$e_i = y_i - \widehat{y}_i$	e_i^2
	14	2	3.4043	-1.4043	1.9721
	16	5	3.8918	1.1082	1.2281
	27	7	6.5731	0.4269	0.1822
	42	9	10.2295	-1.2295	1.5116
	39	10	9.4982	0.5018	0.2518
	50	13	12.1795	0.8205	0.6732
	83	20	20.2235	-0.2235	0.0500
$\sum$	272	66	66.0000	0.00	5.8690

Derivation of regression parameters

- The error of the ith observation

$$e_i = y_i - \widehat{y}_i = y_i - (b_0 + b_1 x_i)$$

- The mean error for a sample of n observations

$$\overline{e} = \frac{1}{n}\sum_{i=1}^{n} e_i = \frac{1}{n}\sum_{i=1}^{n} \{y_i - (b_0 + b_1 x_i)\} = \overline{y} - b_0 - b_1 \overline{x}$$

- Setting mean error to zero

$$b_0 = \overline{y} - b_1\overline{x}$$

- Substitute b0 in the error expression

$$e_i = y_i - \overline{y} + b_1\overline{x} - b_1x_i = \left(y_i - \overline{y}\right) - b_1\left(x_i - \overline{x}\right)$$

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left\{(y_i - \overline{y})^2 + 2b_1(y_i - \overline{y})(x_i - \overline{x}) + b_1^2(x_i - \overline{x})^2\right\}$$

$$\frac{SSE}{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2 - 2b_1\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})(x_i - \overline{x}) + b_1^2\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

$$= s_y^2 - 2b_1s_{xy}^2 + b_1^2s_x^2$$

- Divided by n-1 to facilitate the last notation (n-1 is not related to the degrees of freedom

$$\frac{1}{n-1}\frac{d(SSE)}{db_1} = -2s_{xy}^2 + 2b_1s_x^2 = 0$$

$$b_1 = \frac{s_{xy}^2}{s_x^2} = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n(\overline{x})^2}$$

Allocation of variation

- Error variance without regression = variance of the response

$$Error = e_i = Observed\,response - Predicted\,response = y_i - \overline{y}$$

- Variance of errors without regression =

$$\frac{1}{n-1}\sum_{i=1}^{n} e_i^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2 = Variance\,of\,y$$

- The sum of squared errors without regression

$$\sum_{i=1}^{n}(y_i - \overline{y})^2$$

- SST = total sum of squares
    - It is a measure of y's variability and is called variation of y

$$SST = \sum_{i=1}^{n}(y_i - \overline{y})^2 = \left(\sum_{i=1}^{n} y_i^2\right) - n\overline{y}^2 = SSY - SS0$$

- SSY is the sum of squares of y (or $\sum y^2$)
- SSO is the sum of squares of $\bar{y}$ and is equal to $n\bar{y}^2$

- The difference between SST and SSE is the sum of squares explained by the regression (SSR)
  SSR = SST - SSE
  SST = SSR + SSE
- Coefficient of determination (the fraction of the variation that is explained determines the goodness of the regression)

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST}$$

- The higher the value of R^2 the better the regression
    - $R^2 = 1$ => perfect fit
    - $R^2 = 0$ => no fit

$$Sample\,correlation(x, y) = R_{xy} = \frac{s_{xy}^2}{s_x s_y}$$

$$Coefficient\ of\ determination = \{Correlation\ coefficient(x, y)\}^2$$

- Shortcut formula for SSE

$$SSE = \sum y^2 - b_0 \sum y - b_1 \sum xy$$

Example

- Consider the disk I/O – CPU time data from the previous example

$$SSE = \sum y^2 - b_0 \sum y - b_1 \sum xy$$
$$= 828 + 0.0083 \times 66 - 0.2438 \times 3375 = 5.87$$
$$SST = SSY - SS0 = \sum y^2 - n(\overline{y})^2$$
$$= 828 - 7 \times (9.43)^2 = 205.71$$
$$SSR = SST - SSE = 205.71 - 5.87 = 199.84$$
$$R^2 = \frac{SSR}{SST} = \frac{199.84}{205.71} = 0.9715$$

- The regression explains 97% of the CPU time's variation

Standard deviation of errors

- Variance of errors
  - Divide sum of squared errors by its degrees of freedom
  - Errors are obtained after calculating 2 regression parameters => n-2 degrees of freedom

$$s_e = \sqrt{\frac{SSE}{n-2}}$$

- MSE (mean squared errors) = SSE/(n-2)
- S_e (standard deviation of errors) = sqrt(MSE)
- SSY - n degrees of freedom
  - Obtained from n independent observations without estimating any parameters
- SSO - 1 degree of freedom
  - Computer simply from $\bar{y}$
- SST - n-1 degrees of freedom
  - Only requires one parameter to be calculated from the data $\bar{y}$
- SSR (= SST – SSE) has the remaining 1 degree of freedom

$$SST = SSY - SS0 = SSR + SSE$$
$$(n-1) = n - 1 = 1 + (n-2)$$

- Degrees of freedom add up (similar to sums of squares)

Example

- Consider the disk I/O - CPU time data example
- Degrees of freedom of the sums

$$\begin{array}{lrcrcrcr} SS: & SST & = & SSY & - & SS0 & = & SSR & + SSE \\ & 205.71 & = & 828 & - & 622.29 & = & 199.84 & + 5.87 \\ DF: & 6 & = & 7 & - & 1 & = & 1 & + 5 \end{array}$$

- Mean squared error

$$MSE = \frac{SSE}{DF\ for\ errors} = \frac{5.87}{5} = 1.17$$

- Standard deviation of error

$$s_e = \sqrt{MSE} = \sqrt{1.17} = 1.08$$

CI for regression parameters

- Regression coefficients b0 and b1 are estimates from a single sample of size n
  - => random
  - => using another sample, the estimates may be different
- Consider β0 and β1 the true parameters of the population

$$y = \beta_0 + \beta_1 x$$

- Computed coefficients b0 and b1 are estimates of βo and β1

$$s_{b_0} = s_e \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{b_1} = \frac{s_e}{\left[\sum x^2 - n\overline{x}^2\right]^{\frac{1}{2}}}$$

- The 100(1-α)% confidence intervals for b0 and b1 can be computed using the 1-α/2 quantile of a t variate with n-2 degrees of freedom t[1—α/2; n-2]
- The confidence intervals

$$b_0 \mp t s_{b_0}$$

$$b_1 \mp t s_{b_1}$$

- If a confidence interval includes zero => the regression parameter cannot be considered different from zero at the 100(1-α)% CI

Example

- Consider the I/O – CPU time data example

$$n = 7,\ \overline{x} = 38.71,\ \sum x^2 = 13.855,\ s_e = 1.0834$$

$$s_{b_0} = s_e\left([\frac{1}{n} + \frac{\overline{x}^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$= 1.0834\left[\frac{1}{7} + \frac{(38.71)^2}{13.855 - 7 \times 38.71 \times 38.71}\right]^{\frac{1}{2}} = 0.8311$$

$$s_{b_1} = \frac{s_e}{\left[\sum x^2 - n\overline{x}^2\right]^{\frac{1}{2}}}$$

$$= \frac{1.0834}{[13.855 - 7 \times 38.71 \times 38.71]^{\frac{1}{2}}} = 0.0187$$

$$b_0 \mp ts_{b_0} \qquad b_1 \mp ts_{b_1} \qquad t_{[1-\alpha/2;\ n-2]} = t_{[0.95;\ 5]}$$

- 90% confidence interval for b0

$$-0.0083 \mp (2.015)(0.8311) = (-1.6830, 1.6663)$$

- CI includes zero - the hypothesis that this parameter is zero cannot be rejected at 0.10 significance level => b0 is essentially zero
- 90% confidence interval for b1

$$0.2438 \mp (2.015)(0.0187) = (0.2061, 0.2814)$$

- CI does not include zero - the slope b1 is significantly different from zero at this confidence level

Example

- Best linear models
  Time on UNIX = 0.017 (Data size in bytes) + 24
  Time on ARGUS = 0.034 (Data size in bytes) +30
- The regression explains 81% and 75% of the variation, respectively.
  Does ARGUS take longer time per byte than UNIX?
  Does ARGUS have a longer setup time than UNIX?

- Intervals for intercepts overlap, while those for slopes do not
- Setup times are not significantly different in the 2 systems, while the per byte times (slopes) are different

UNIX			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	26.898	2.005	(23.2968, 30.4988)
b1	0.017	0.003	(0.0128, 0.0219)
ARGUS			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	31.068	4.711	(22.6076, 39.5278)
b1	0.034	0.006	(0.0231, 0.0443)

CI for predictions

- Mean value of the predicted response

$$\hat{y}_p = b_0 + b_1 x_p$$

- Standard deviation of the mean of a future sample of m observations

$$s_{\hat{y}_{mp}} = s_e \left[\frac{1}{m} + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

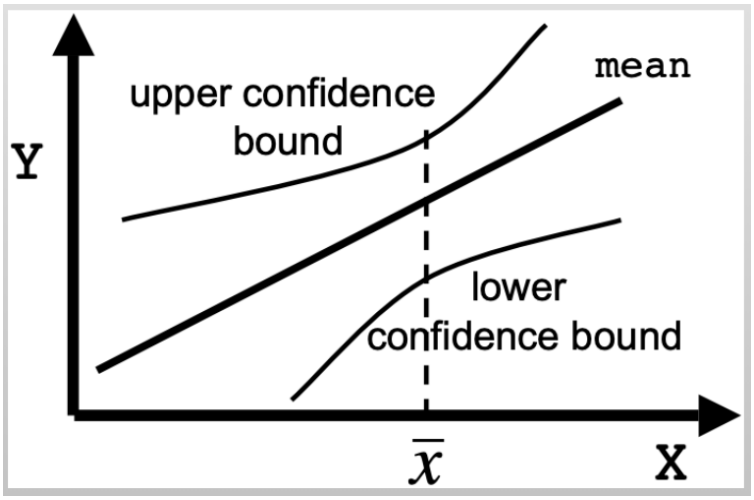
- m = 1 => standard deviation of a single observation

$$s_{\hat{y}_{1p}} = s_e \left[1 + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

- m = ∞ => standard deviation of the mean of a large number of future observations at xp

$$s_{\hat{y}_p} = s_e \left[\frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

- 100(1-α)% confidence interval for the mean can be constructed using a t quantile read at n-2 degrees of freedom
- Goodness of the prediction decreases as we move away from the center



Example

- Consider the disk I/O - CPU time data example
  CPU time = -0.0083 + 0.2438(Number of disk I/O's)
- What is the mean CPU time for a program with 100 disk I/O's?
  CPU time = -0.0083 + 0.2438(100) = 24.3674
- Standard deviation of errors S_e = 1.0834
- The standard deviation of the predicted mean of a large number of observations

$$s_{\widehat{y}_p} = s_e\left[\frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{\widehat{y}_p} = 1.0834\left[\frac{1}{7} + \frac{(100 - 38.71)^2}{13855 - 7(38.71)^2}\right]^{\frac{1}{2}} = 1.2159$$

- 90% CI for the predicted mean

$$24.3574 \mp (2.015)(1.2159) = (21.9174, 26.8174)$$

- CPU time of a single future program with 100 disk I/Os

$$s_{\widehat{y}_{1p}} = s_e\left[1 + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{\widehat{y}_p} = 1.0834\left[1 + \frac{1}{7} + \frac{(100 - 38.71)^2}{13855 - 7(38.71)^2}\right]^{\frac{1}{2}} = 1.6286$$

- 90% CI for a single prediction

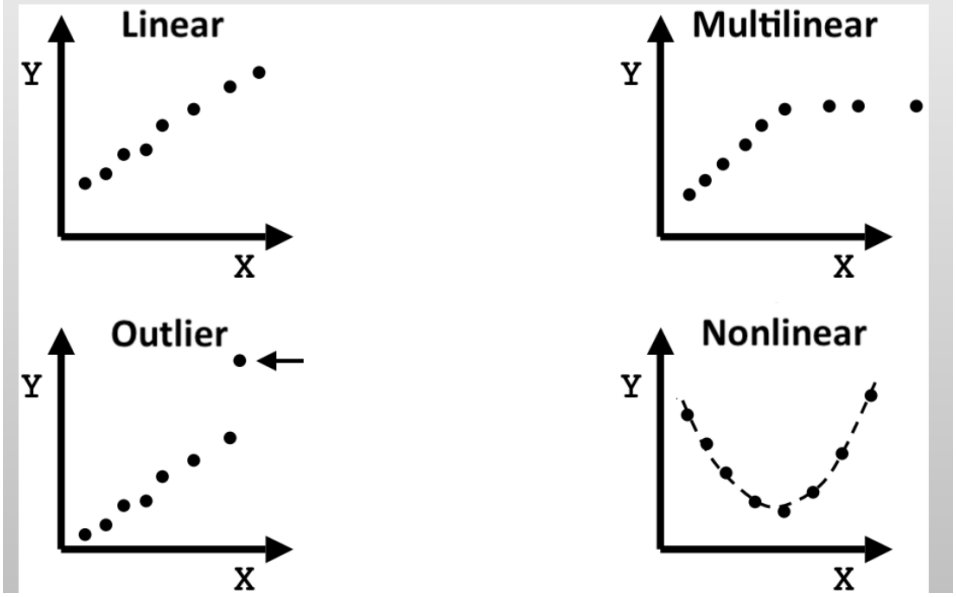
$$24.3574 \mp (2.015)(1.6286) = (21.0858, 27.6489)$$

Visual tests for regression assumptions

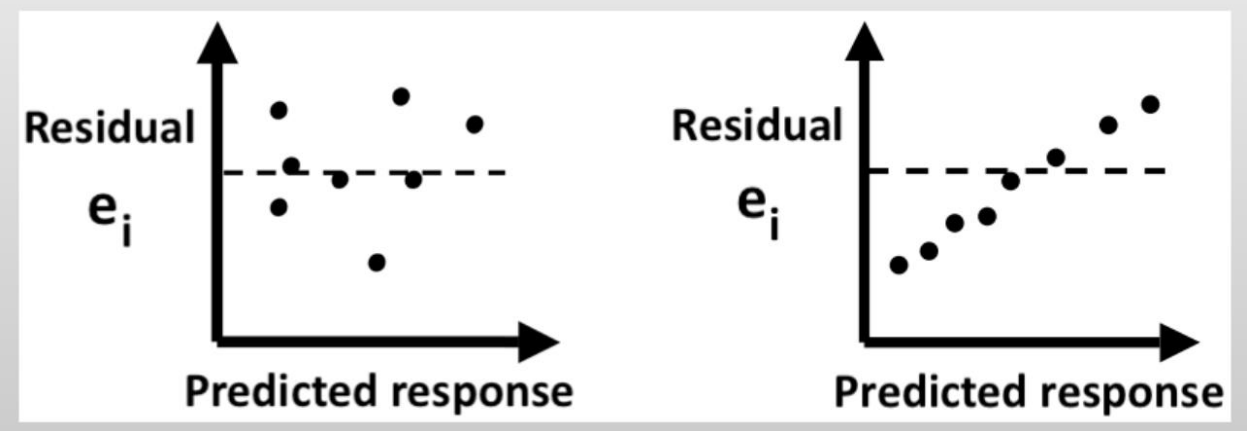
Regression assumptions

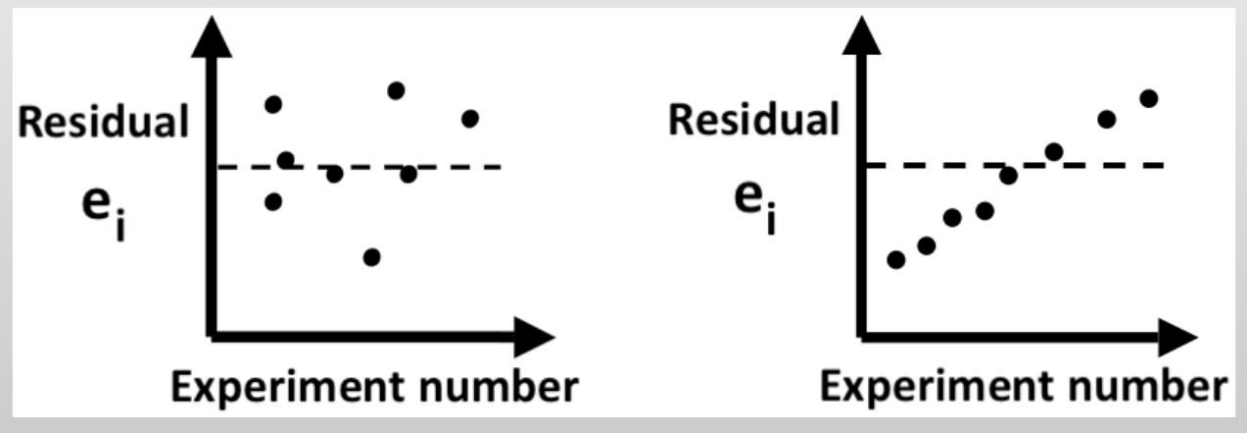
1. The true relationship between the response variable y and the predictor variable x is linear
2. The predictor variable x is non-stochastic and it is measured without any error
3. The model errors are statistically independent
4. The errors are normally distributed with zero mean and a constant standard deviation

1. Linear relationship



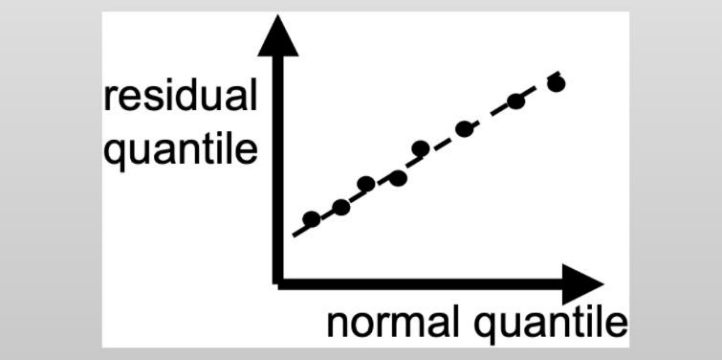
2. Independent errors



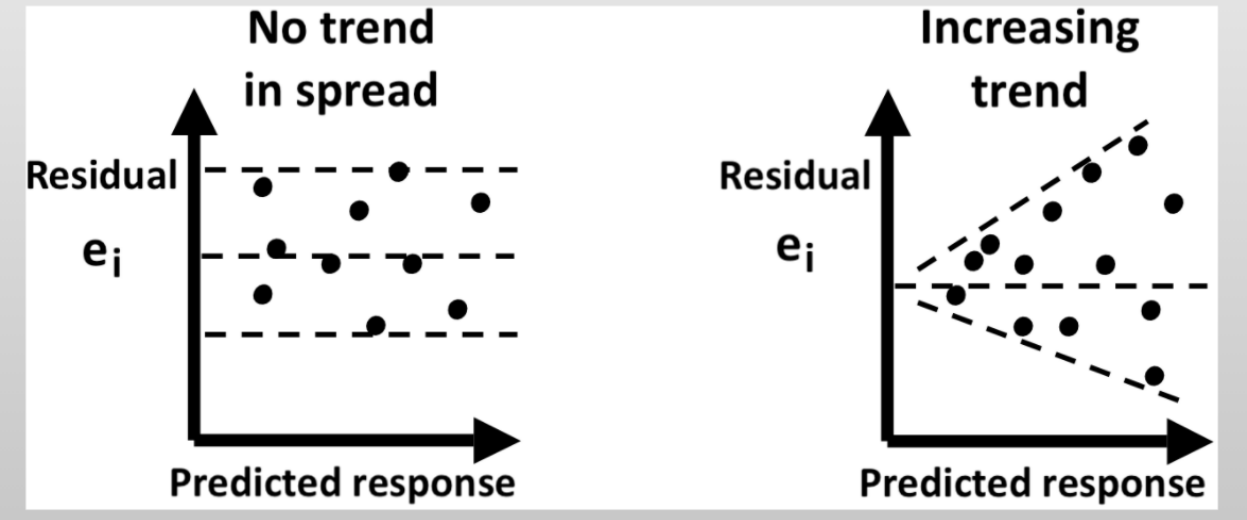


3. Normally distributed errors

- Use a quantile-quantile plot of e_i versus N(0,1)
    - Linear => the assumption is satisfied

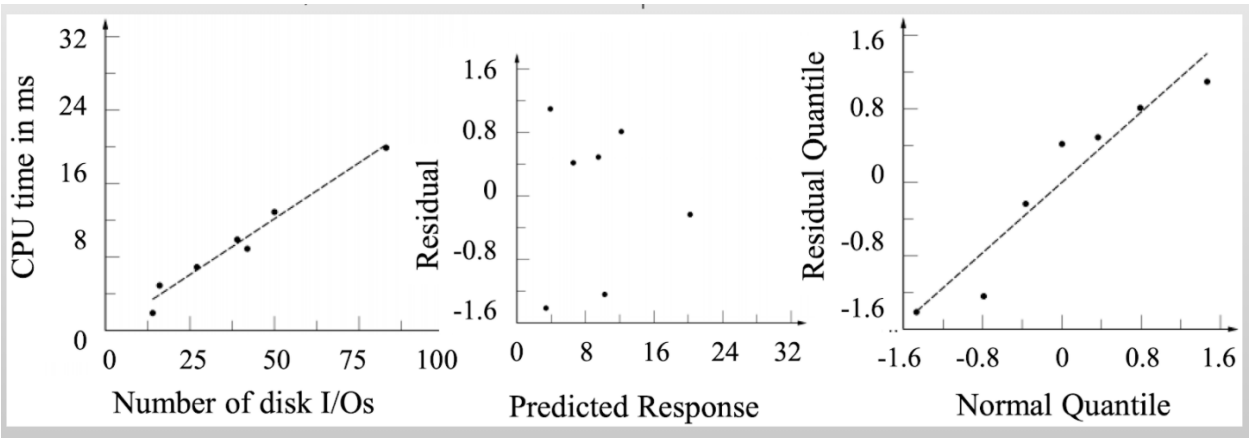


4.Constant standard deviation of errors



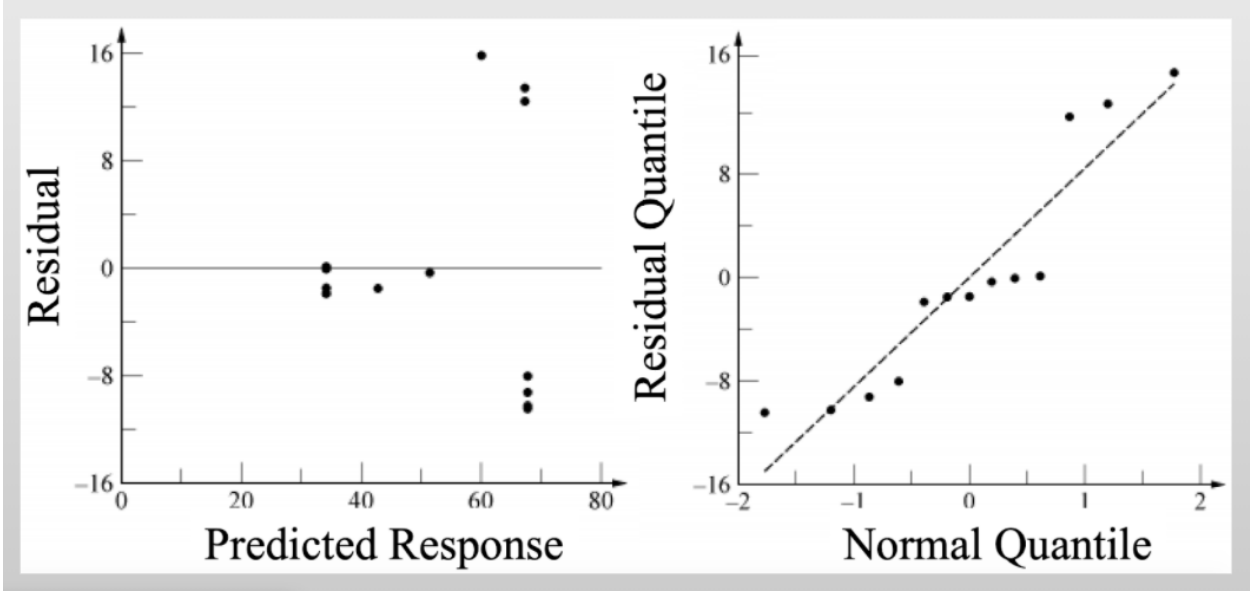
Example

Consider the disk I/O – CPU time data example



1. Relationship is linear
2. No trend in residuals seem independent
3. Linear normal quantile-quantile plot
    - Larger deviations at lower values but all values are small

Example



1. Large errors at larger responses
2. Normality of errors is questionable